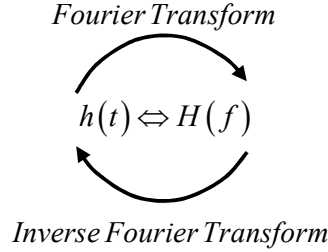


Fourier Transformation

1. Basic Definition and Uses Fourier Transformation

- (a) It is a principle analysis tool in many of today's scientific challenges.
 - (b) Fourier series is discrete summation whereas Fourier transform is continuous.
 - (c) It is the most well-known application of the mathematical technique to analyze linear time-invariant systems.
 - (c) It is a universal problem-solving technique.
 - (d) In Fourier transformation simultaneous visualization of function and its Fourier transform is a key for analysis.
- ⇒ Fourier transform has many definitions and this is chosen according to visibility.

Suppose we have a function, Then



$h(t)$ and $H(F)$ are known as Fourier transformation pairs.

Uses:

1. Image processing is based on Fourier transformation.
2. Fourier transform is used in solving partial differential equations in electromagnetic theory.
3. Fourier transform is also used in Quantum Mechanics.

Basic definitions of Fourier transform:

$$H(f) = \int_{-\infty}^{\infty} h(t) e^{-j2\pi ft} dt : \text{Fourier Transform}$$

$$h(t) = \int_{-\infty}^{\infty} H(f) e^{j2\pi ft} df : \text{Inverse Fourier transform}$$

$$\Rightarrow H(f) = R(f) + jI(f), \text{ Where}$$

- $H(f)$ is a complex function.
- $R(f)$ is a real part of $H(f)$.
- $I(f)$ is the imaginary part of $H(f)$.

$$\Rightarrow |H(f)| = \sqrt{R^2 + I^2} \Rightarrow |\theta(f)| = \tan^{-1} \frac{I}{R}$$

Some other useful definitions:

In terms of $\omega = 2\pi f$

$$H(\omega) = a_1 \int_{-\infty}^{\infty} h(t) e^{-j\omega t} dt \quad \& \quad h(t) = a_2 \int_{-\infty}^{\infty} H(\omega) e^{-j\omega t} d\omega$$

Where a_1 & a_2 can be $\Rightarrow a_1 = a_2 = 1$

$$\Rightarrow a_1 = \frac{1}{2\pi}, a_2 = 1$$

$$\Rightarrow a_1 = 1, \quad a_2 = \frac{1}{2\pi}$$

$$\Rightarrow a_1 = \frac{1}{\sqrt{2\pi}} \quad a_2 = \frac{1}{\sqrt{2\pi}}$$

$\Rightarrow$ Time and frequency are reciprocal to each other.

$\Rightarrow$ Position and wave vector are reciprocal to each other.

Definition of Fourier Transformation in position (r) and wave vector (k) space:

$$F(k) = \int_{-\infty}^{\infty} f(x) e^{ikx} dx$$

$$f(x) = \int_{-\infty}^{\infty} F(k) e^{-ikx} dk$$